

Supplementary Material S2

MMAT Appraisal by Study and Domain

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder:
A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the figure in the manuscript presenting the aggregate distribution of MMAT judgements by domain (Section 2.6). Here the appraisal is reported study by study.

In accordance with the manual of the Mixed Methods Appraisal Tool (MMAT) version 2018, which deliberately eliminated the numerical summative scoring present in earlier versions, no aggregate score per study is computed or reported. Each study was assigned to the corresponding design category and appraised with the five criteria proper to that category.

#	Reference	Type	Cat.	Rep.	Meas.	Part.	Conf.	Out.	ML base	Group
1	Johnson et al. (2025)	Quant. non-randomised	3	Y	Y	Y	CT	Y	Eligible	
2	He et al. (2019)	Quant. descriptive	4	Y	Y	Y	Y	Y	Eligible	
3	Reed et al. (2021)	Quant. descriptive	4	Y	Y	Y	Y	Y	Eligible	
4	Lecciso et al. (2021)	Mixed methods	5	Y	Y	Y	CT	Y	Eligible	
5	Shaik et al. (2025)	Quant. non-randomised	3	CT	Y	CT	N	Y	Eligible	
6	Ahmad et al. (2024)	Quant. non-randomised	3	N	Y	N	N	Y	Eligible	
7	El Rhatassi et al. (2024)	Quant. non-randomised	3	CT	Y	CT	N	Y	Contextual ref.	
8	Radočaj & Martinović (2025)	Quant. non-randomised	3	Y	Y	Y	Y	Y	Eligible	
9	Washington et al. (2022)	Quant. non-randomised	3	Y	Y	Y	Y	Y	Eligible	
10	Sholikah et al. (2024)	Quant. non-randomised	3	CT	Y	CT	N	Y	Y	Eligible
11	Levy et al. (2024)	Systems design	–	–	–	–	–	–	–	Contextual ref.
12	Sariyanidi et al. (2025)	Quant. non-randomised	3	Y	Y	Y	Y	Y	Y	Eligible
13	Hassan et al. (2025)	Quant. non-randomised	3	Y	Y	Y	Y	Y	Y	Eligible
14	Alsubai (2025)	Quant. non-randomised	3	N	Y	N	N	Y	–	Eligible

(Continued)

#	Reference	Type		Cat.	Rep.	Meas.	Part.	Conf.	Out.	ML base	Group
15	Rudovic et al. (2018)	Quant. non-randomised		3	Y	Y	Y	Y	Y	Y	Eligible
16	Cicceri et al. (2026)	Quant. non-randomised		3	Y	Y	Y	CT	Y	Y	Eligible
17	Rudovic et al. (2019)	Quant. non-randomised		3	Y	Y	Y	Y	Y	Y	Eligible
18	Almeida et al. (2019)	Systems design		—	—	—	—	—	—	—	Eligible
19	Alam et al. (2025)	Quant. non-randomised		3	CT	Y	N	N	Y	Y	Eligible
20	Hwang et al. (2024)	Quant. non-randomised		3	Y	Y	Y	Y	Y	Y	Eligible
21	Aljbori et al. (2025)	Quant. non-randomised		3	N	Y	N	N	Y	Y	Eligible
22	El Rhatassi et al. (2025)	Quant. non-randomised		3	CT	Y	CT	N	Y	Y	Contextual ref.
23	Ramachandran et al. (2025)	Quant. non-randomised		3	CT	Y	CT	N	Y	Y	Eligible
24	Hosney et al. (2024)	Quant. non-randomised		3	N	Y	N	N	Y	Y	Eligible
25	Nagae & Lee (2022)	Quant. descriptive		4	Y	Y	Y	CT	Y	—	Contextual ref.
26	Papaioannou et al. (2021)	Quant. descriptive		4	Y	Y	Y	Y	Y	—	Contextual ref.
27	Dapogny et al. (2019)	Quant. descriptive		4	Y	Y	Y	Y	Y	Y	Eligible
28	Leo et al. (2018)	Quant. non-randomised		3	Y	Y	Y	Y	Y	Y	Eligible
29	Akin-Bulbul et al. (2024)	Quant. non-randomised		3	Y	Y	Y	CT	Y	Y	Eligible
30	Xu & Ju (2023)	Quant. descriptive		4	Y	Y	Y	Y	Y	—	Contextual ref.
31	O'Sullivan et al. (2023)	Quant. non-randomised		3	Y	Y	Y	Y	Y	—	Contextual ref.

Note. Y = Yes; N = No; CT = Cannot Tell. The five appraisal columns are abbreviated as Rep. (sample representativeness), Meas. (appropriateness of measurements), Part. (protocol and sampling strategy), Conf. (confounding control and analysis) and Out. (outcome data completeness); their correspondence with the original MMAT criteria is set out in the table below. These labels are deliberately not numbered D1 to D5, since in the manuscript that notation designates the five clinical domains of the corpus taxonomy, which are a different classification. The Cat. column indicates the MMAT 2018 category applied: 3 (quantitative non-randomised, criteria 3.1–3.5), 4 (quantitative descriptive, criteria 4.1–4.5) and 5 (mixed methods, criteria 5.1–5.5). Studies 11 and 18 correspond to systems designs without supervised classification, a category not contemplated by the instrument, and are marked with a dash in all appraisal columns. The ML base column marks with Y the 21 studies with an identifiable ML architecture oriented to the face or to emotion, which constitute the denominator of the architecture, hyperparameter and class-balancing tables

of the manuscript; the 10 studies marked with a dash comprise 6 clinical or behavioural studies without a computational component (2, 3, 4, 25, 26 and 30), 2 systems designs (11 and 18) and 2 studies whose supervised model operates on non-facial modalities (14 and 31). Membership in this base is independent of the eligibility group: it includes 2 contextual reference studies and excludes 4 eligible ones. The Group column distinguishes the 24 eligible studies from the 7 contextual reference studies defined in Section 2.4 of the manuscript; the correspondence of each study with its clinical domain, dataset and provenance is documented in Supplementary Material S4, which employs the same numbering of the # column. The 3 retracted studies were excluded from the corpus following the editorial status verification described in Section 2.7 of the manuscript and do not receive methodological quality appraisal.

Correspondence of domains with the original MMAT 2018 criteria

Column	Description	Cat. 3	Cat. 4
Rep.	Sample representativeness	3.1	4.2
Meas.	Appropriateness of measurements	3.2	4.3
Part.	Protocol and sampling strategy	3.5	4.1
Conf.	Confounding control and analysis	3.4	4.5
Out.	Outcome data completeness	3.3	4.4

Aggregate distribution by domain

Domain	Cat. 3 (n = 22) Y / N / CT	Cat. 4 (n = 6) Y / N / CT
D1 Sample representativeness	12 / 4 / 6	6 / 0 / 0
D2 Appropriateness of measurements	22 / 0 / 0	6 / 0 / 0
D3 Protocol and sampling strategy	12 / 5 / 5	6 / 0 / 0
D4 Confounding control and analysis	9 / 10 / 3	5 / 0 / 1
D5 Outcome data completeness	22 / 0 / 0	6 / 0 / 0

The single mixed methods study of the corpus (Lecciso et al., 2021, criteria 5.1–5.5) meets four of the five domains, with a cannot-tell judgement in confounding control and analysis. The sum $22 + 6 + 1 + 2$ recovers the 31 studies of the corpus.

Instrument reference

Hong, Q. N., Fàbregues, S., Bartlett, G., Boardman, F., Cargo, M., Dagenais, P., Gagnon, M.-P., Griffiths, F., Nicolau, B., O'Cathain, A., Rousseau, M.-C., Vedel, I., & Pluye, P. (2018). The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information*, 34(4), 285–291. <https://doi.org/10.3233/EFI-180221>